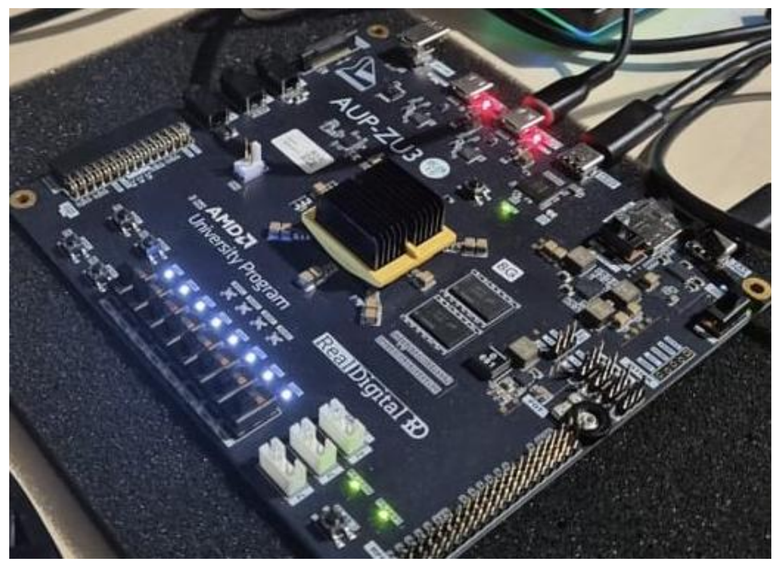
PointNet at 2 to 4 Bits with Brevitas and FINN+: Sensitivity-Guided Mixed Precision on a ZU3EG

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Motivation

- Drones, robots and vehicles must classify what they see in real time, but cannot carry a GPU. Power and size budgets will not allow it.
- PointNet, a neural network architecture for classifying 3D point cloud data, poses particular dataflow-mapping challenges such as per-point shared MLPs, global max pooling, and T-Nets whose batched matrix multiplication lacks a fixed-topology FINN mapping.
- Aggressive quantization enables deployment on a small FPGA, but uniform 2-bit quantization drops ModelNet40 accuracy by 21.53 percentage points. Quantizing each layer to the same width does not cost the same accuracy.
- Research Question: can layer-wise sensitivity guide an efficient mixed-precision accelerator for 3D point cloud classification?

Mapping PointNet to FINN+

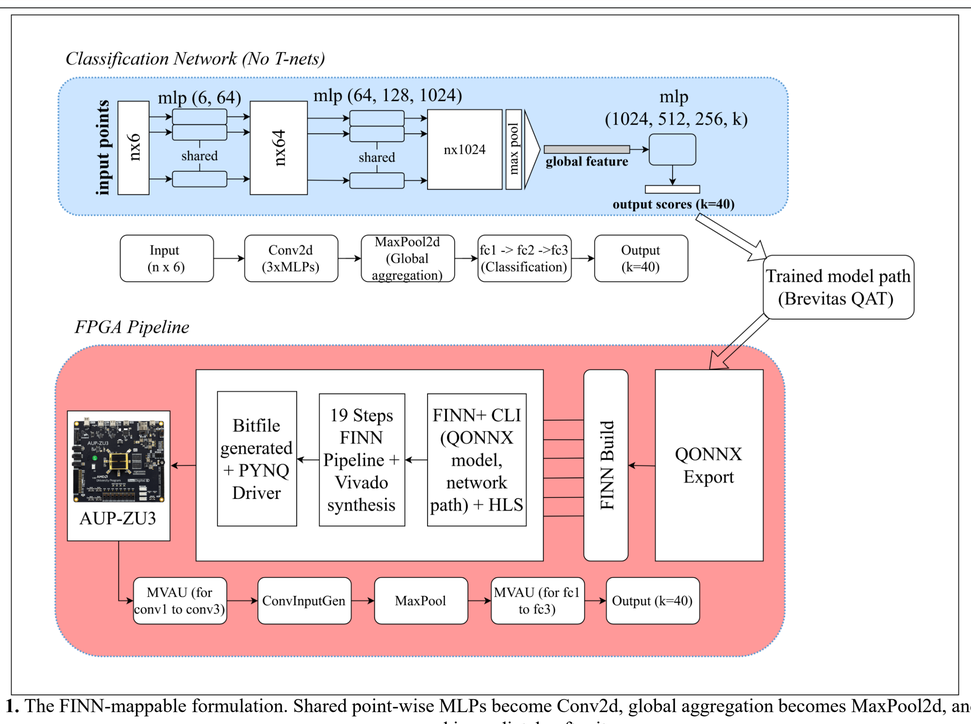


Fig. 1. The FINN-mappable formulation. Shared point-wise MLPs become Conv2d, global aggregation becomes MaxPool2d, and fc1 is exposed immediately after it.

- T-Nets are removed and surface normals supply the six input channels.
- The mapping preserves the pre-aggregation and post-aggregation split that the next section turns on.

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Sensitivity-Guided Mixed Precision

ALGORITHM 1 Layer-wise weight-only PTQ probe

Require: FP32 model $\mathcal{M}$, test set $\mathcal{D}$, layers $\mathcal{L}$, bit-widths $\mathcal{B} = \{2, 3, 4\}$
Ensure: sensitivity scores $S[\ell][b]$

$acc_{base} \leftarrow$ Evaluate($\mathcal{M}$, $\mathcal{D}$)
for each $b \in \mathcal{B}$ do
  for each $\ell \in \mathcal{L}$ do
    $\mathcal{M}' \leftarrow$ Copy($\mathcal{M}$)
    quantize weights of ℓ to b bits;
    all other layers remain FP32
    $S[\ell][b] \leftarrow acc_{base} -$ Evaluate($\mathcal{M}'$, $\mathcal{D}$)
  end for
end for
return S

1. Load the FP32 baseline
2. Select one layer
3. Quantize only that layer to 2, 3 or 4 bits. All others stay FP32
4. Measure the accuracy drop against FP32

a larger drop means higher layer sensitivity

- Six layers at three bit-widths, 18 evaluations in total. No Hessian analysis, no automated search.

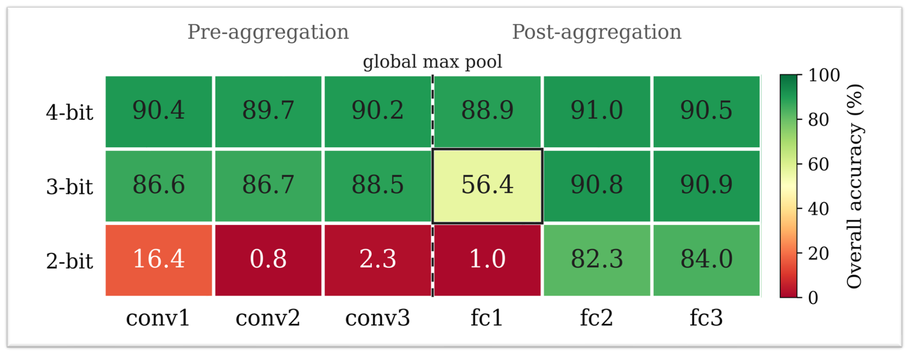


Fig. 2. One layer quantized to the stated bit-width, every other layer left at FP32. Baseline 91.1% on ModelNet40.

- Max-pooling keeps only the largest value in each of the 1,024 channels and discards the rest, so rounding error in a point that is not the maximum never reaches the output. fc1 sees the pooled vector with no such filter, and nothing downstream can repair it. Every layer before the pool recovers by 3 bits. fc1 does not, and it alone sets the precision the whole network must carry.
- conv3 and fc2 hold identical weight counts (131,072) on opposite sides of the boundary, yet score 2.3% and 82.3% at 2 bits. Parameter count does not explain sensitivity. Position does.
- PTQ is diagnostic, not predictive. The conv layers collapse under 2-bit PTQ yet train to useful accuracy under QAT. The probe ranks layers by vulnerability to quantization; it does not forecast final accuracy.

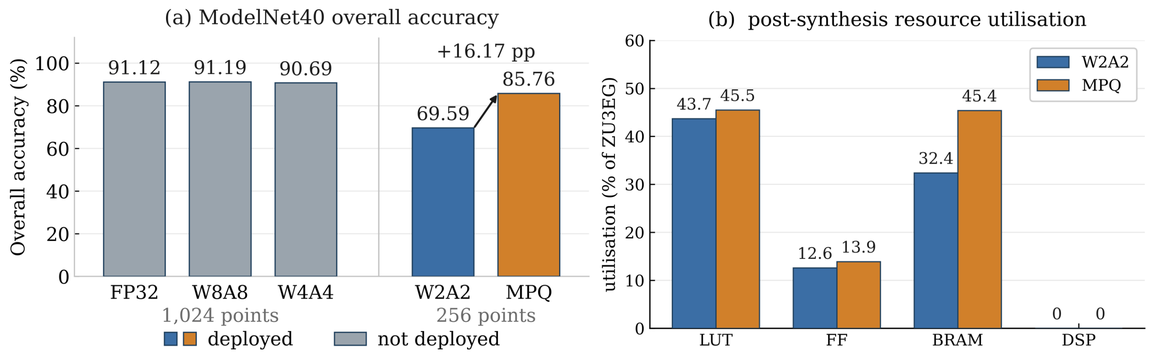
Engineering decision: W4A4 at fc1, W2A2 elsewhere.

FPGA Deployment Results

2,663	1.25 ms	45.5%	0
FPS	Latency	LUT Utilization	DSPs

Measured on board with hardware cycle counters over a 10 second run, accelerator only, excluding host DMA.

- The sensitivity-guided allocation changes only fc1 from W2A2 to W4A4; all other layers and FINN+ folding remain unchanged.
- This isolates the hardware effect of adding precision at the identified bottleneck.



- Selective precision recovers most of what uniform-2 bit loses: MPQ gains 16.17 percentage points over uniform W2A2 while leaving five of six quantized layers at 2 bits.
- Higher fc1 precision is a resource trade, not a timing trade, because both deployed models retain the same pipeline bottleneck, so throughput and latency remain unchanged.
- The added cost is mainly memory-side: fc1's 4-bit weight matrix raises BRAM by 40% while LUTs rise only 4%. All MACs remain DSP-free.

Work	Bits	Points	Latency	OA (%)	Platform
Bai et al.	8/16	4,096	10.9 ms	N/R	ZCU104
BiPointNet	1	1,024	9.0 ms	85.62%	RPi 4B
This work	**2 to 4**	**256**	**1.25 ms**	**85.76%**	**ZU3EG**

Table 1. Comparison with prior point-cloud FPGA and embedded deployments.

- The ZU3EG deployment reaches 1.25 ms latency on a substantially smaller device, showing that the low-bit FINN+ dataflow remains practical under a tight LUT budget.
- Prior systems use different point counts and platforms, so the comparison demonstrates deployment scale rather than a controlled architectural speedup.

Key Takeaways

- PointNet quantization sensitivity is structured around the global aggregation boundary.
- The first fully-connected layer after aggregation is the critical precision bottleneck.
- Layer-wise accuracy-drop analysis guides effective allocation without Hessian analysis or automated search. PTQ ranks layers; it does not predict final accuracy.
- Raising only fc1 from 2-bit to 4-bit recovers 16.17 percentage points of accuracy over uniform 2-bit, costs 1,304 extra LUTs (under 2% of the device), and runs at the same speed.

Future Work

- Move the sliding-window buffer to BRAM. It currently consumes 13,250 LUTs, 41.3% of the design, in distributed RAM.
- Add on-board power measurements to quantify energy efficiency and report FPS/W.
- Extend the sensitivity probe beyond PointNet to aggregation-based point-cloud networks such as PointNet++ and DGCNN.
- Restore the T-Nets with a FINN-compatible batch matrix multiplication operator, enabling evaluation of the full PointNet architecture.

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